Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18MEDE723]

# SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2021

## AUTOMIBILE ENGINEERING

**Time: 3 Hours** **Maximum Marks: 70**

**Instructions:**
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

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|---|---|---|---|
| **1.** | **a.** | Explain the various types of cylinder arrangements used in the multi cylinder engine. | **06** |
| | **b.** | Explain the working principle of turbo charging and mention the limitations of turbocharging. | **08** |
| **2.** | **a.** | With a neat sketch, explain the working of single plate clutch. | **06** |
| | **b.** | A motor vehicle has a wheel base of 2.75 m and pivot centers at a distance of 1.05 m apart. The front & rear wheel track is 1.25 m. Determine the correct angle of outside lock and turning circle radius of outer front and inner rear wheels when angle of inside lock is 40°. | **08** |
| **3.** | **a.** | Explain with help of suitable sketch, the construction of the disc wheel. | **06** |
| | **b.** | Discuss in details, the various factors affecting tyres life. | **08** |
| | | **OR** | |
| **4.** | **a.** | What is a fuel cell? Explain the working principle of fuel cell. | **06** |
| | **b.** | Discuss the concept of the hybrid vehicle with a neat sketch. | **08** |
| **5.** | **a.** | With a neat sketch, explain motion of vehicle along curved path. | **06** |
| | **b.** | An automobile has a wheel base of 2.65 m and height of C.G above the ground is 0.6 m and it is 1.13 m in front of rear axle. If automobile is moving with a speed of 45 km/hr, find minimum stopping distance when:<br>**i.** only rear wheels are braked<br>**ii.** only front wheels are braked | **08** |
| | | **OR** | |
| **6.** | **a.** | Explain the working of traction control system used in passenger cars. | **06** |
| | **b.** | Briefly describe the construction & working of hydraulic braking system. | **08** |
| **7.** | **a.** | Explain the sources of automotive exhaust gas emission. | **06** |
| | **b.** | Explain the controlling of crank case emissions with a neat sketch. | **08** |
| | | **OR** | |
| **8.** | **a.** | Discuss in detail the greenhouse effect & global warming potential. | **06** |
| | **b.** | Elaborate the causes, effects and control measures of the noise pollution caused by automobiles. | **08** |